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graceful
Biology & AI
brilliant observant outstanding youthful adaptive

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Group 8 - No.2

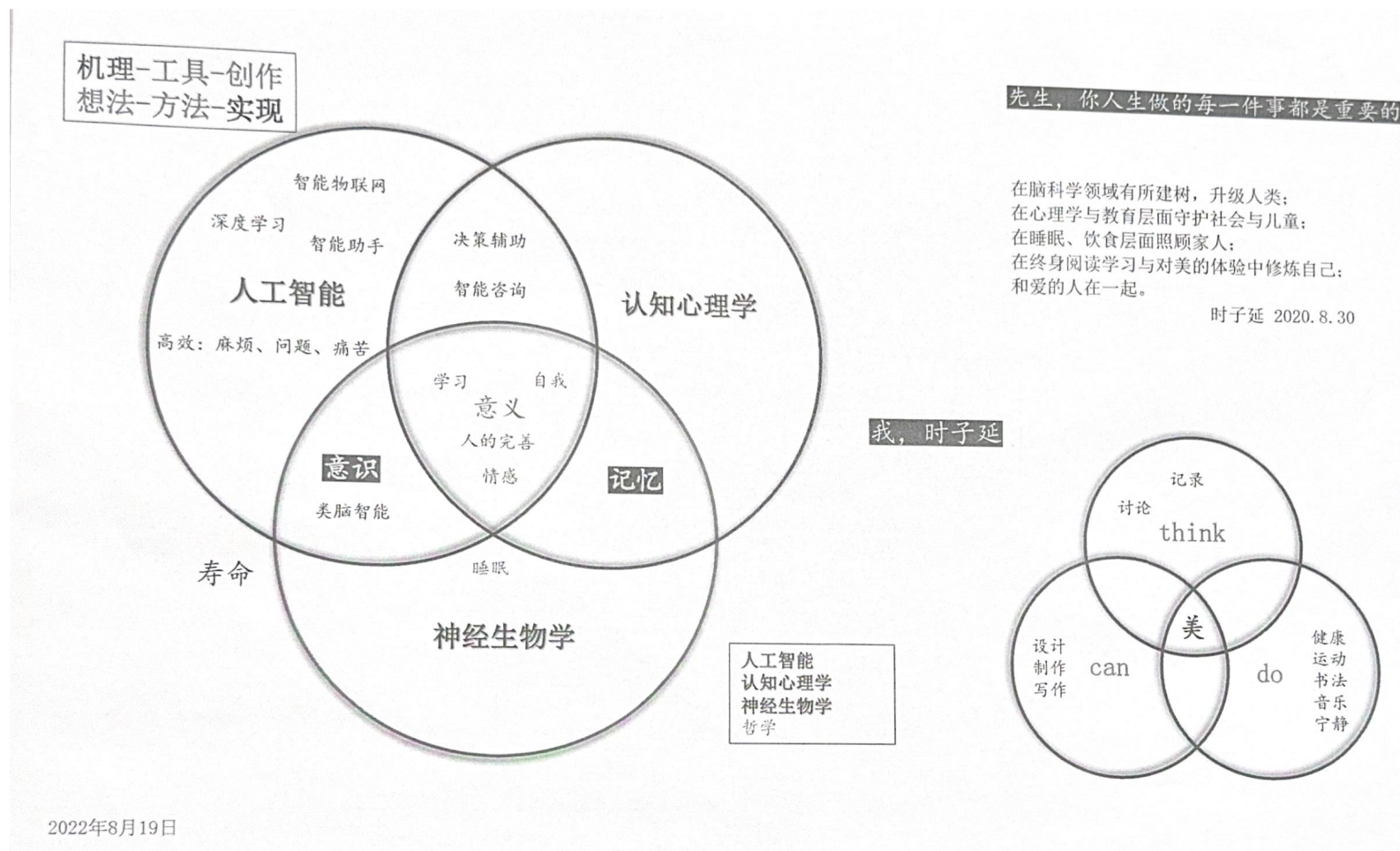
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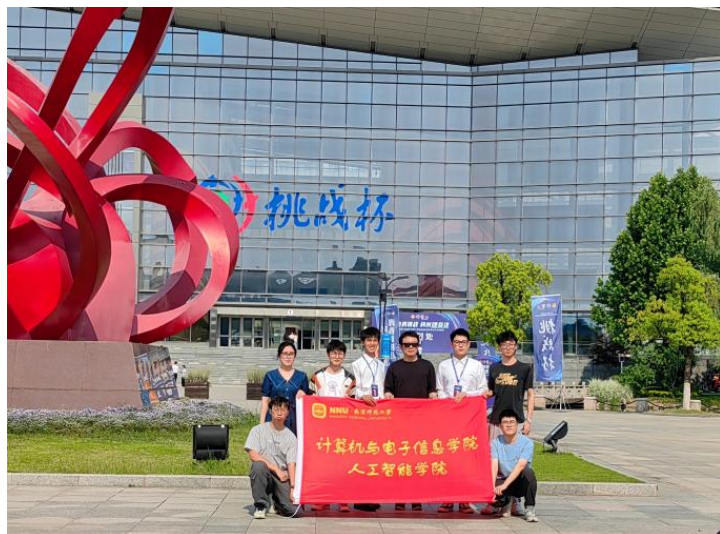
Research Experience

From AI for Science to Brain-inspired Computing

Research Plan

Brain-inspired mRAG and fTHz-based BCI





1. **2025.** Z. Shi, Y. Gu, H. Cui, et al. *From Machine Learning to Multimodal Models: The AI Revolution in Enzyme Engineering*. **BioDesign Research**, 2025. doi: [10.1016/j.bidere.2025.100044](https://doi.org/10.1016/j.bidere.2025.100044). (SCI, IF 4.7, JCR Q1)
2. **2024.** S. Yuan, N. Peng, Z. Shi, et al. *Detection of algal tiny objects based on morphological features*. **Concurrency Computat Pract Exper**, 2024. doi: [10.1002/cpe.8128](https://doi.org/10.1002/cpe.8128). (SCI, JCR Q2)
3. **2023.** S. Yuan, N. Peng, Z. Shi, et al. *Magic: A Morphable Attention Based Algal Tiny Object Detection Model*. **IEEE BESC**, 2023. doi: [10.1109/BESC59560.2023.10386562](https://doi.org/10.1109/BESC59560.2023.10386562). (EI)

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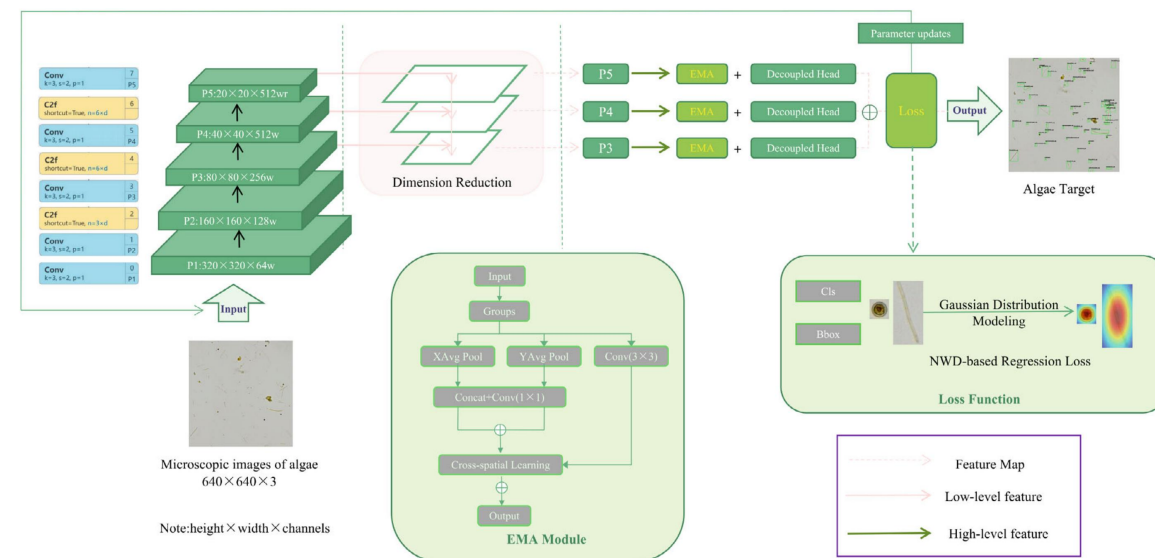


FIGURE 1 The framework of morphable attention based algal tiny object detection strategy.

							
aphanocapsa_sp	centricae_sp	chlorella_sp	chlorophyta_3	cosmarium_sp	cryptomonas_sp	cyanophyta_sp	cyclotella_sp
							
cylindrospermopsis_cuspid		cylindrospermopsis_gangetica		cylindrospermopsis_raciborskii		dactylococcopsis_acicularis	
							
limnithrix_obliqueacuminata		limnithrix_sp	microcystis_1	navicula_sp	pennatae_sp	planktolyngbya_limnetica	
							
pseudanabaena_sp		raphidiopsis_sinensis	rhabdoglea_smithii	rhabdoglea_sp	scenedesmus_sp	ulnarina_acus	

FIGURE 6 Examples of algae label mentioned.



2023 Conference

MAGic: A Morphable Attention Based Algal Tiny Object Detection Model

Shuai Yuan ^{1*}, Ningkang Peng ^{1*}, **Ziyan Shi** ¹, Yanhui Gu ¹, Beili Sun ^{2,3}
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Abstract—The dense and toxic blooms formed by cyanobacteria in the aquatic environment have caused serious damage to public health and aquatic ecosystems. It is crucial to monitor and prevent cyanobacterial blooms in freshwater waters in a timely manner. Although the object detection method has been applied to the field of algae identification, there are some issues to be solved in the existing research. The main challenge is that the setting of the training set is too idealistic, which is separated from the actual water quality environment and cannot guide the actual algae identification and water quality monitoring. In this paper, we collect 2,024 algae microscopic images of algae from a reservoir in southern China and a data set for object detection. Aiming at the above problems, we proposed a morphable attention based algal tiny object detection strategy to sample and extract features of algae small targets more effectively. It is hoped that the model training and improvement results of this paper can help to establish a future algae real-time monitoring and water quality evaluation system.

Index Terms—cyanobacterial blooms, water quality evaluation, algal detection, object detection, computer vision

I. INTRODUCTION

Cyanobacteria (Cyanobacteria algae) can proliferate in freshwater and marine environments to form dense and toxic blooms[1, 2]. In recent years, the outbreak of harmful cyanobacterial blooms in freshwater lakes and rivers has intensified worldwide. China faces the same problem, for example, cyanobacteria blooms every year in Dianchi Lake and Lake Taihu. The excessive growth and reproduction of cyanobacteria have caused serious damage to public drinking water sources and aquatic ecosystems[3]. Therefore, it is critical to monitor and control cyanobacterial blooms in freshwater bodies[4, 5].

The species and quantity of cyanobacteria in water samples are one of the important indicators for analyzing and judging the severity of water blooms, and also provide guidance for the prevention and control of subsequent water blooms[4]. The traditional algae detection method is artificial microscopy, which uses an optical microscope to qualitatively and quantitatively analyze phytoplankton. However, the disadvantages of the method are also obvious. The number of algae classification counts reaches hundreds of thousands in water environmental monitoring, while the manual microscopy method is time-consuming, labor-intensive and costly along with missed or false detections[6]. With the continuous progress and application of machine learning technology, the research findings in the field of computer vision have been migrated to the

field of algae detection[7–9]. The image classifier performs well in the field of phytoplankton identification[3, 10–12]. At present, a variety of object detection frameworks have been applied to algae identification[13–15], which replaces the traditional microscopy method and improves the efficiency of classification and identification[16–19]. For example, Cao et al.[16]used the improved YOLOv3 model of MobileNet[20] and SPP[21] to achieve better recognition performance on two single-category microalgae datasets, indicating that the object detection model can use the image features of algae not rich for recognition. Park et al.[18]applied the YOLOv3 model with Darknet53[22] as the backbone to train on a data set with a limited number of algae, which proved that the lightweight object detection model has more advantages in such tasks. Zhou et al.[19]investigated variants of TOOD[23], YOLOv5[24], YOLOv8[25] and RCNN[26] algorithms to train on artificially set algae imbalanced data sets, which proved that the high accuracy performance of existing object detection models on such tasks.

However, there are some issues to be solved in the current research literature. The main challenge is that the setting of the training set is too idealized, such as too few categories and unnatural sampling sources, disconnection from the actual water quality environment, and inability to describe real algae indicators. Therefore, even if a good training result is obtained, it cannot guide the actual algae identification and water quality monitoring. In this paper, we collected 2,024 microscopic images of algae from a reservoir in Zhuhai, including 31,919 targets of 80 algae genera. On this basis, we create a data set for object detection. However, we find that the performance of classical or advanced object detection methods is not ideal on this real data set. The main reason is that with the increase of the sample algae category, the morphological effects between different algae targets are more obvious, including the interference of similar forms and the sampling loss of some small targets. It is a common feature of the large-scale data of all algae datasets, corresponding to the coexistence of many different algae in the water body. We hope to improve the applicability of object detection technology in complex algae recognition tasks by solving this common problem.

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RESEARCH ARTICLE

WILEY

Detection of algal tiny objects based on morphological features

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Summary

The dense and toxic blooms formed by cyanobacteria in aquatic environments pose significant threats to public health and aquatic ecosystems. Timely monitoring and prevention of cyanobacterial blooms in freshwater bodies are thus imperative. Although object detection methods have been applied in the field of algae identification, existing research faces several challenges. A primary issue is the overly idealistic setting of training sets, which are disconnected from the actual water quality environments, impeding practical algae identification and water quality monitoring. In this paper, we collect 2024 microscopic images of algae from a reservoir in Southern China, forming a comprehensive and diverse dataset for object detection. Addressing the aforementioned challenges, we propose an attention-based strategy for the detection of tiny algal objects, which effectively samples and extracts features from algal targets. Our model uniquely leverages the morphable characteristics of algae, enhancing the accuracy and efficiency of identification. The training and improvement results of this model presented in our study are expected to aid in establishing a future system for real-time algae monitoring and water quality assessment.

KEYWORDS

algal detection, computer vision, cyanobacterial blooms, object detection, water quality evaluation

1 | INTRODUCTION

Cyanobacteria (Cyanobacteria algae) can proliferate in freshwater and marine environments to form dense and toxic blooms.^{1,2} In recent years, the outbreak of harmful cyanobacterial blooms in freshwater lakes and rivers has intensified worldwide. China faces the same problem, for example, cyanobacteria blooms every year in Dianchi Lake and Lake Taihu. The excessive growth and reproduction of cyanobacteria have caused serious damage to public drinking water sources and aquatic ecosystems.³ Therefore, it is critical to monitor and control cyanobacterial blooms in freshwater bodies.^{4,5}

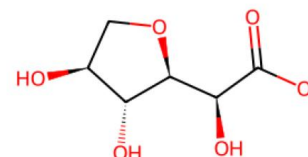
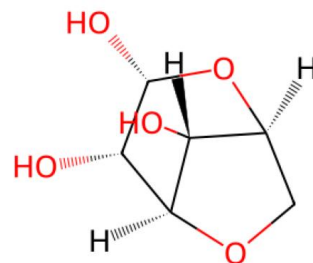
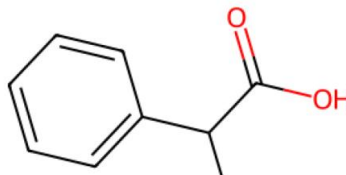
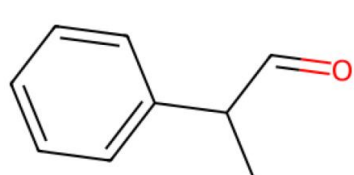
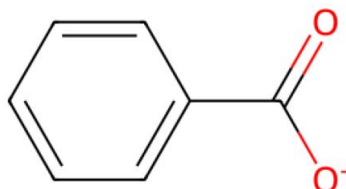
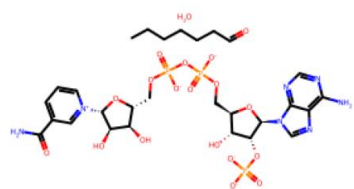
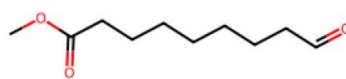
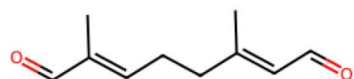
The species and quantity of cyanobacteria in water samples are one of the important indicators for analyzing and judging the severity of water blooms, and also provide guidance for the prevention and control of subsequent water blooms.⁵ The traditional algae detection method is artificial microscopy, which uses an optical microscope to qualitatively and quantitatively analyze algae. However, the disadvantages of the method are also obvious. The number of algae classification counts reaches hundreds of thousands in water environmental monitoring, while the manual microscopy method is time-consuming, labor-intensive and costly along with missed or false detections.⁶ With the continuous progress and application of machine learning technology, a lot of manual work can be taken by algorithms and achieve good results.^{7,8} Logically, the research findings in the field of computer vision have been migrated to the field of algae detection.^{10–12} The image classifier performs well in the field of algae identification.^{3,13–15} At present, a variety of object detection frameworks have been applied to algae identification,^{16–18} which replaces the

iGEM & Hackathon





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Outline

Abstract

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2. Functional Evaluation and Metrics

3. From Individual-Enzyme Function Modelin...

4. Modeling of Enzyme Structure, Properties,...

5. Datasets

6. AI Modeling Paradigms

7. Tasks on Enzyme Engineering

8. Challenge and Outlook

9. Conclusion

Author Contributions

Declaration of Competing Interest

Acknowledgments

References



BioDesign Research

Available online 29 August 2025, 100044

In Press, Journal Pre-proof

What's this?



From Machine Learning to Multimodal Models: The AI Revolution in Enzyme Engineering

Ziyan Shi^{1,2,3,†}, Shuping Xu^{1,2,†}, Sihan Xue^{1,2,†}, Kaiming Chen^{1,4}, Yifan Lu¹, Feiyue Wang⁵, Dr. Siyu Long⁶, Dr. Yannan Tian^{7,8}, Dr. Peng Zhang⁹, Dr. Jianing Wang⁹, Dr. Yanhui Gu³, Dr. Junsheng Zhou³, Dr. Hao Zhou⁶, Dr. Shuaiqi Meng^{1,2}✉, Dr. Haiyang Cui^{1,2}✉

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Review Article

From machine learning to multimodal models: The AI revolution in enzyme engineering

Ziyan Shi^{a,b,c,1}, Shuping Xu^{a,b,1}, Sihan Xue^{a,b,1}, Kaiming Chen^{a,d}, Yifan Lu^a, Feiyue Wang^e, Siyu Long^f, Yannan Tian^{g,h}, Peng Zhangⁱ, Jianing Wang^j, Yanhui Gu^k, Junsheng Zhou^k, Hao Zhou^l, Shuaiqi Meng^{a,b,*,}, Haiyang Cui^{a,b,*,}

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^b Ministry of Education Key Laboratory of NSLSCS, Nanjing Normal University, Nanjing, 210097, China
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^g Department of Systems Biology, School of Life Sciences, Southern University of Science and Technology, No. 1088 Xueyuan Avenue, Shenzhen, 518055, China
^h Institute for Biological Electron-Microscopy, Southern University of Science and Technology, No. 1088 Xueyuan Avenue, Shenzhen, 518055, Guangdong, China
ⁱ State Key Laboratory of Microbial Technology, Shandong University, No. 72 Bihai Road, Qingdao, Shandong, 266237, China

ABSTRACT

Protein engineering is a powerful tool for applications spanning synthetic biology, biocatalysis, and drug discovery. Recent advances in artificial intelligence (AI), from conventional machine learning (ML) algorithms to large-scale pre-trained protein models, have greatly accelerated enzyme engineering field entering a data-driven era. This review provides a guidance map of current enzyme engineering tasks and builds an integrative perspective on AI methods, model types, landmark tasks, and data resources. We begin by delineating the core modeling tasks in enzyme engineering, which include encompassing function annotation, structural modeling, and property prediction and by reviewing recent advances alongside dominant algorithmic frameworks. Next, we outlined the evolution of AI integration into enzyme engineering, which can be broadly categorized into four stages: classical machine learning approaches, deep neural networks, protein language models (pLMs), and emerging multimodal architectures. Finally, we highlight four converging trends that are redefining the landscape of AI-driven enzyme design: (i) the replacement of handcrafted features with unified, token-level embeddings; (ii) a shift from single-modal models toward multimodal, multitask systems; (iii) the emergence of intelligent agents capable of reasoning; and (iv) a movement beyond static structure prediction toward dynamic simulation of enzyme function. Together, these developments are paving the way for intelligent, generalizable, and mechanistically interpretable AI platforms poised to synthetic biology.

1. Introduction

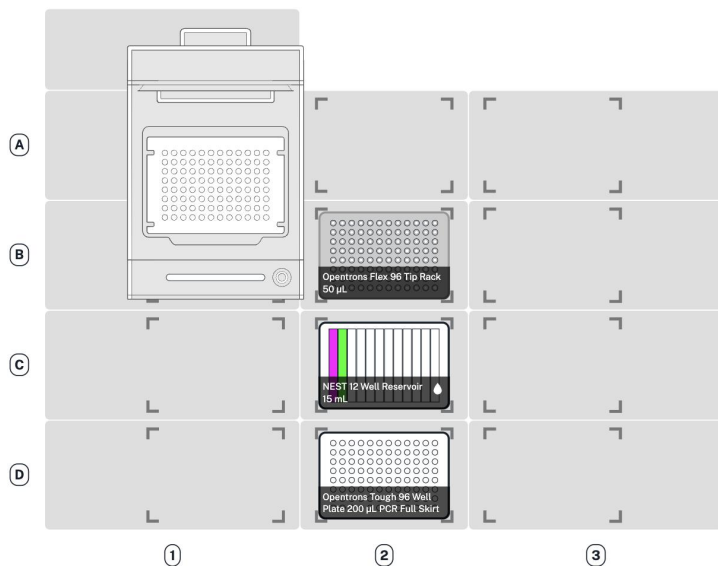
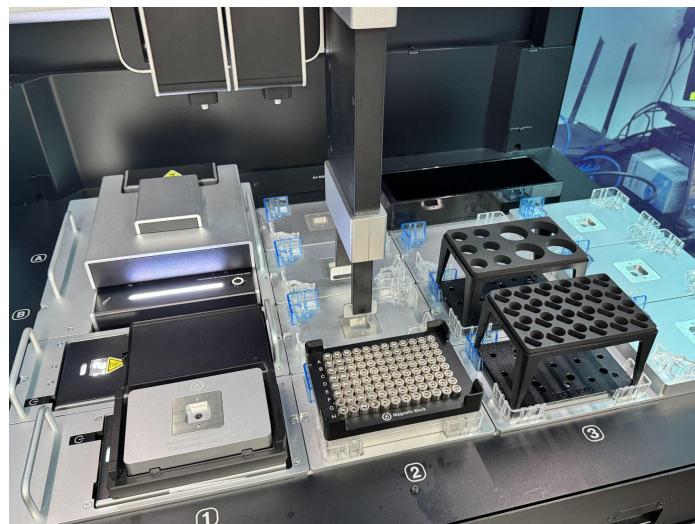
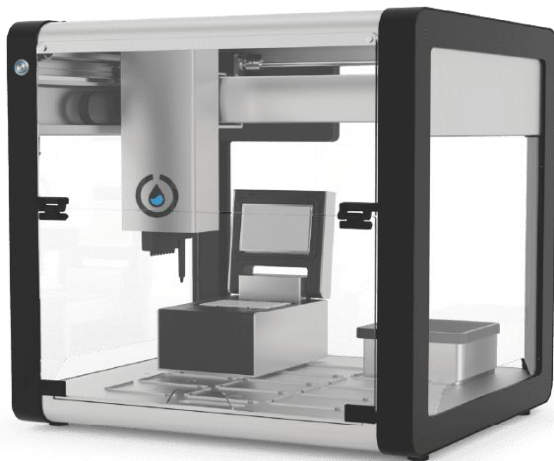
1.1. The central role of enzymes

Enzymes are an important class of biomolecules with catalytic functions, playing indispensable roles in sustaining life [1]. They are essential not only in biological systems but also in a wide array of industries applications [2]. Enzymes are predominantly proteins produced by living cells, with a small fraction being catalytic RNAs. Enzyme often exhibit high substrate specificity and remarkable catalytic efficiency, making them indispensable in various biological and industrial processes. Due to their structural diversity and functional versatility, enzymes hold substantial application value in areas such as synthetic biology, metabolic engineering, and therapeutic development. In enzymatic reactions, substrates bind to enzymes to form enzyme-substrate complexes, resulting in stabilized interactions and reduced molecular degrees of freedom. The stabilization of the transition-state complex and the associated entropy reduction lower the activation energy, thereby significantly enhancing the reaction rate. A prevailing model for substrate recognition is the induced fit theory, which posits that substrate binding induces conformational changes in the enzyme. This structural rearrangement allows the enzyme to reshape its active site to accommodate the ligand more precisely, facilitating the formation of the enzyme-substrate complex and promoting efficient catalysis.

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Note: Low-resolution images were used to create this PDF. The original images will be used in the final composition.



```
NUM_SAMPLES = 8 # 样本数 (1~96)
REACTION_VOL = 10.0 # µL, 总体积
TEMPLATE_VOL = 1.0 # µL
ENZYME_VOL = 5.0 # µL 绿色rapid taq
PRIMER_F_VOL = 1.0 # µL (已按原0.25µL做了1:4稀释)
PRIMER_R_VOL = 1.0 # µL (已按原0.25µL做了1:4稀释)
WATER_VOL = 2.0 # µL (配合引物稀释后调整)
```

```
# =====
# Step 4: 分装 Master Mix 到新PCR孔 (或在原孔上做PCR也可)
# 这里选择在同一块板的“目标PCR孔位”中装反应体系, 以免裂解残留影响体积精度。
# 演示: 把目标PCR孔位为 sample_wells 对应位置 (也可以换到新列/新板)
# =====
protocol.comment("Step 4: 向目标PCR孔分装 9 µL Master Mix.")
target_wells = sample_wells # 如需换到新列/新板, 请修改此行

# 使用单通道逐孔分配 (样本少时稳妥; 样本多可用8通道逐列分配)
for w in target_wells:
    p50_1.pick_up_tip()
    p50_1.aspirate(per_rxn_mm, mm_tube.bottom(2))
    p50_1.dispense(per_rxn_mm, w.bottom(DISPENSE_HEIGHT_OFFSET))
    p50_1.blow_out(w.top()) # 轻吹以确保全量
    p50_1.return_tip()

# =====
# Step 5: 每孔加入 1 µL 模板 (来自前述裂解&中和后的样本孔)
# 这里假设模板就来自 sample_wells (同孔位), 为了干净, 模板吸取后直接打入 target_wells。
# =====
protocol.comment("Step 5: 每孔加入 1 µL 模板。")
for src, dest in zip(sample_wells, target_wells):
    p50_1.pick_up_tip()
    p50_1.aspirate(TEMPLATE_VOL, src.bottom(0.5))
    p50_1.dispense(TEMPLATE_VOL, dest.bottom(1.0))
    p50_1.mix(2, 6, dest) # 轻微混匀
    p50_1.return_tip()

# =====
# Step 6: 上机PCR
# =====
tc.open_lid()
protocol.move_labware(labware=pcr_plate, new_location=tc, use_gripper=True)
tc.close_lid()
tc.set_lid_temperature(105)

# 初始变性
tc.set_block_temperature(**initial_denaturation)

# 循环步骤
tc.execute_profile(steps=cycle_steps, repetitions=cycle_number, block_max_volume=REACTION_VOL)

# 末端延伸
tc.set_block_temperature(**final_extension)

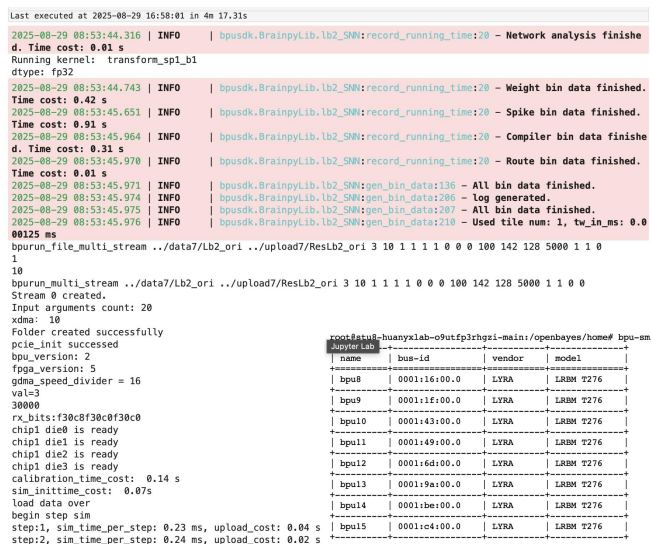
# 结束保温
tc.set_block_temperature(hold_temp)
tc.open_lid()
protocol.comment("PCR 完成。")
```



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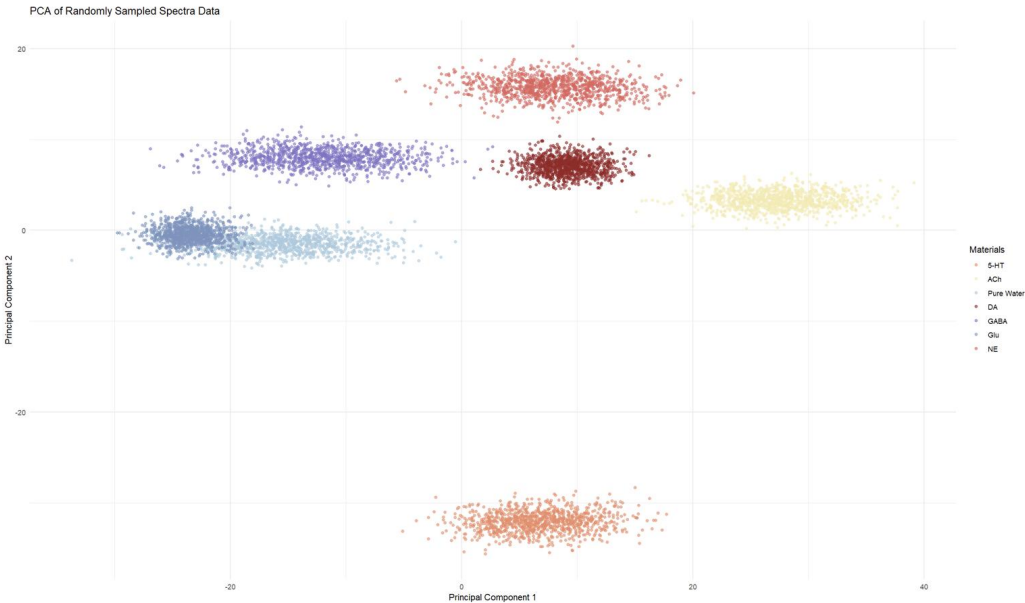
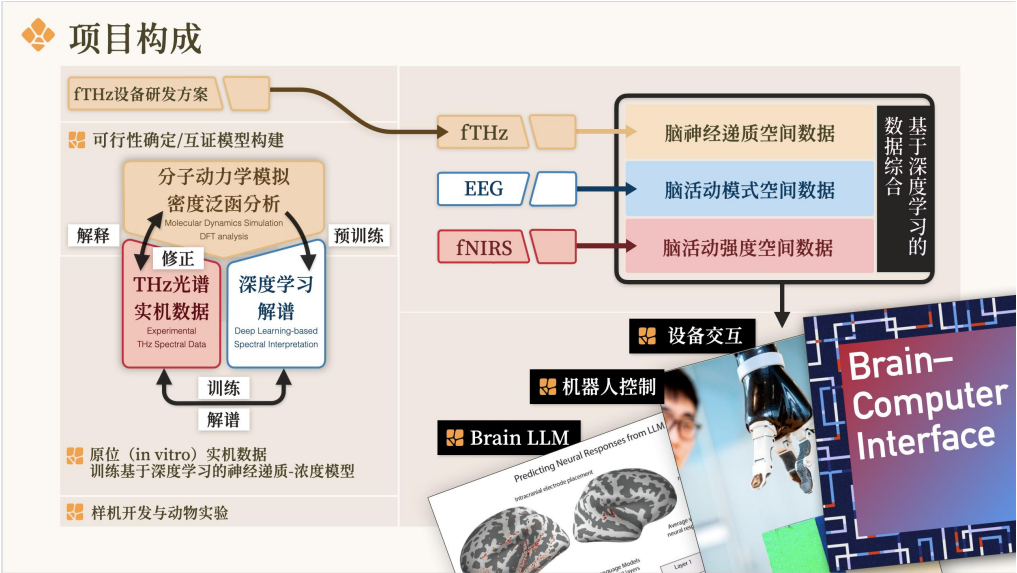
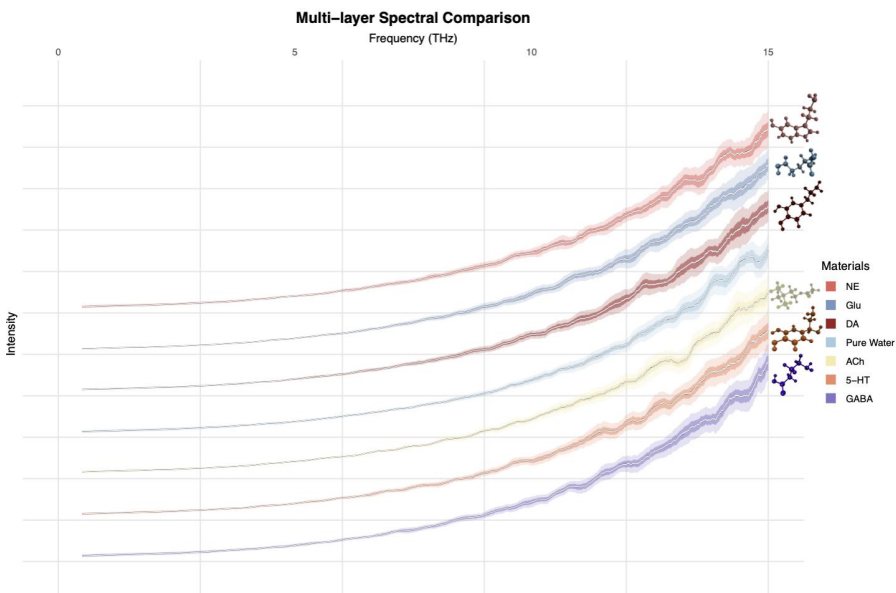
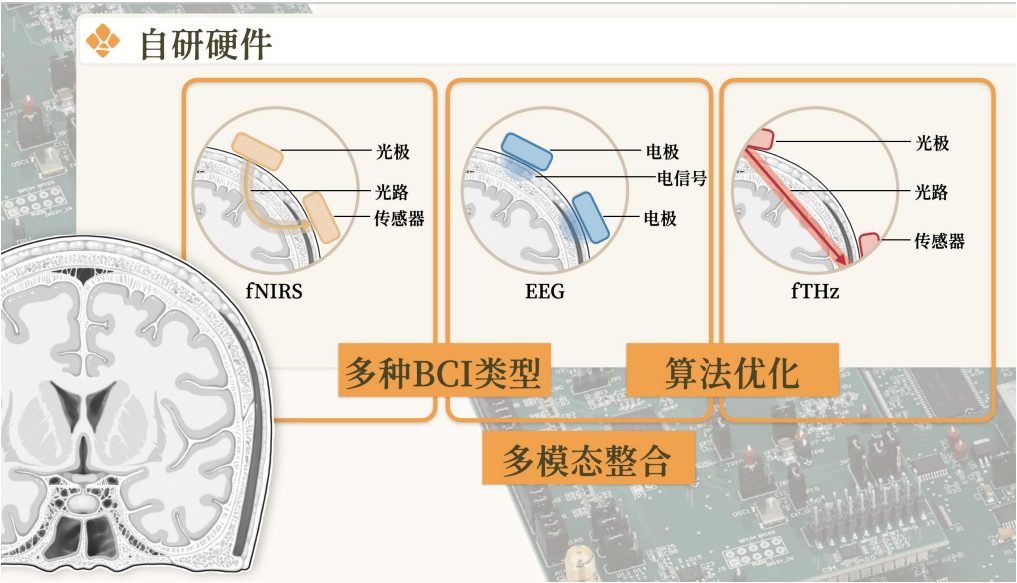
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A woman with dark hair, wearing a black sleeveless top and a patterned skirt, is speaking into a microphone at a podium. She is smiling and looking towards the audience. The background is a plain wall.

Host

Invited







专利(1项)

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发明人: 彭宁康 姚佳怡 时子延 顾彦慧



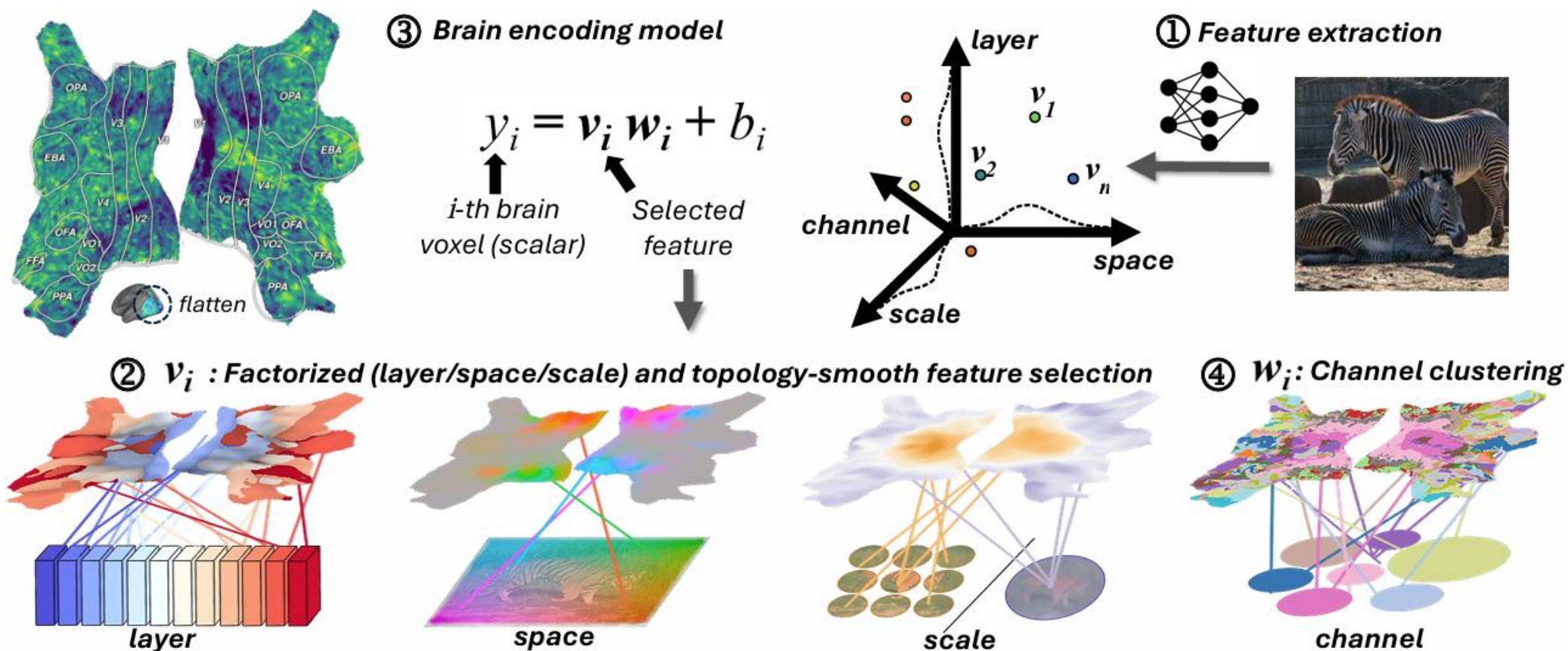
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➤ Brain for AI

➤ Brain Decodes deep Nets(CVPR2024)



- Brain-Inspired mRAG Retriever: From Intent Inference to Expected Visual Embedding
 - Predict Before You Retrieve
 - Intent Recognition
 - Task-Gated Fusion
 - Expected Visual Embedding (EVE)
 - Goal-Oriented Retrieval
 - Surprise-Driven Learning
 - Replay (Sleep-Like Consolidation)
- Predict Cofactor using CLEAN
- BCI - YuEEG (8 channels)



Thanks for your time.

Q&A